

Intermediate Level Probability Notes

1. Operational Properties of Random Variables

Expectation of Sums: $E[X + Y] = E[X] + E[Y]$. This property is called linearity of expectation and does not require independence. Variance of a Scaled Variable: $\text{Var}(aX) = a^2 \text{Var}(X)$. Variance measures spread, and scaling a variable stretches the spread by the square of the constant. Variance Definition: $\text{Var}(X) = E[X^2] - (E[X])^2$.

2. Joint Probability Distributions

Joint PMF (Discrete Case): $p(x, y) = P(X = x, Y = y)$. Total probability condition: $\sum_x \sum_y p(x, y) = 1$. Joint PDF (Continuous Case): $\int \int f(x, y) dx dy = 1$. Joint distribution models the probability of co-occurrence of two random variables.

3. Marginal Distributions

Marginal Distribution (Discrete): $p(X = x) = \sum_y p(x, y)$. Marginal Distribution (Continuous): $f_X(x) = \int f(x, y) dy$. $f_Y(y) = \int f(x, y) dx$. Marginalization means summing or integrating out the other variable.

4. Conditional Probability and Density

Conditional Density: $f(y | x) = f(x, y) / f(x)$, provided $f(x)$ is not zero. Conditional Expectation: $E[X | Y]$ represents the expected value of X when Y is fixed. Law of Iterated Expectation: $E[X] = E[E[X | Y]]$.

5. Covariance and Uncorrelated Variables

Covariance Definition: $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$. Equivalent Formula: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$. If $\text{Cov}(X, Y) = 0$, then X and Y are uncorrelated. If X and Y are independent, then: $E[XY] = E[X]E[Y]$, which implies $\text{Cov}(X, Y) = 0$. However, uncorrelated does not always imply independence.